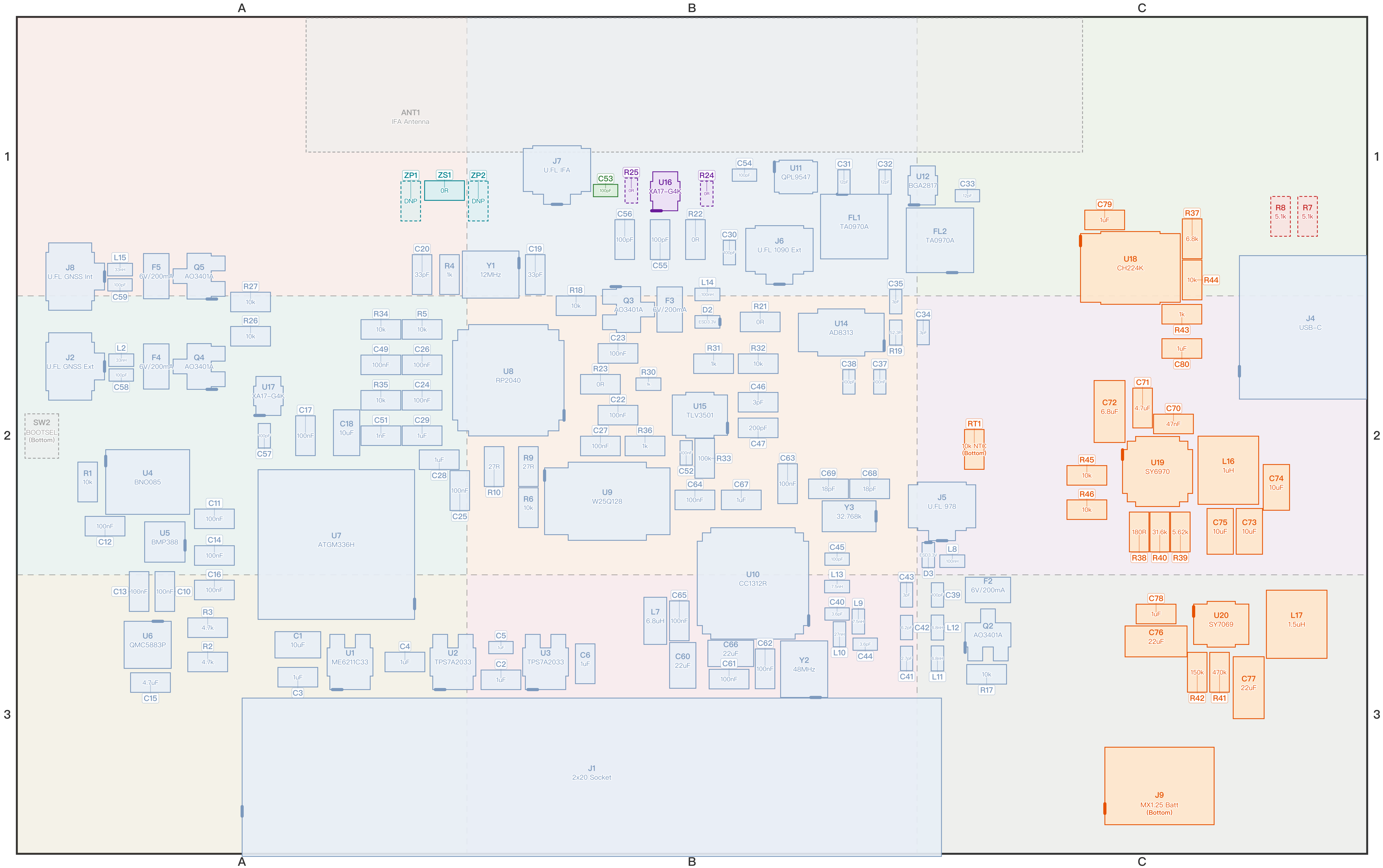
Pilot Kit Expansion Board V4.0 Assembly Map — Powered Variant (Top View)

Parts are drawn to their real outline and orientation; value inside the outline, designator outside. A heavy | marks pin 1 / the polarised end. The nine background tints (columns A/B/C, rows 1/2/3) match the grid badges on page 2. Board 100.0×62.0mm, scaled 5.62×.

Standard 131 · Power Section 28 · No-Power Variant 2 · Antenna Select 3 · IFA Match 3 · Debug Position 1 · No Part 2 · Dashed = not fitted 8



170 designators (excluding H1–H4 mounting holes and FID1–3 fiducials); 8 of them are not fitted on this variant. | Placement notes on page 3 | Rules: SELECTIVE_PLACEMENT-zh_CN.md

Pilot Kit Expansion Board V4.0 Designator Index — Grid Reference

Each grid badge is the same colour as its cell on page 1 — match the colour to jump straight to it. Flags: ✕ = not fitted, \| = polarised (line it up with the polarity mark on page 1). Designators marked ※ have a specific note on page 3.

Ref	Grid	Type	Flags	Value	Footprint
ANT1※	A1	No Part	✕	IFA Antenna	PCB copper
C1	A3	Standard		10uF	0805
C2	B3	Standard		1uF	0603
C3	A3	Standard		1uF	0603
C4	A3	Standard		1uF	0603
C5	B3	Standard		1uF	0402
C6	B3	Standard		1uF	0603
C10	A3	Standard		100nF	0603
C11	A2	Standard		100nF	0603
C12	A2	Standard		100nF	0603
C13	A3	Standard		100nF	0603
C14	A2	Standard		100nF	0603
C15	A3	Standard		4.7uF	0603
C16	A3	Standard		100nF	0603
C17	A2	Standard		100nF	0603
C18	A2	Standard		10uF	0805
C19	B1	Standard		33pF	0603
C20	A1	Standard		33pF	0603
C22	B2	Standard		100nF	0603
C23	B2	Standard		100nF	0603
C24	A2	Standard		100nF	0603
C25	A2	Standard		100nF	0603
C26	A2	Standard		100nF	0603
C27	B2	Standard		100nF	0603
C28	A2	Standard		1uF	0603
C29	A2	Standard		1uF	0603
C30	B1	Standard		100pF	0402
C31	B1	Standard		12pF	0402
C32	B1	Standard		12pF	0402
C33	C1	Standard		12pF	0402
C34	C2	Standard		3pF	0402
C35	B2	Standard		3pF	0402
C37	B2	Standard		100nF	0402
C38	B2	Standard		100pF	0402
C39	C3	Standard		100pF	0402
C40	B3	Standard		3.6pF	0402
C41	B3	Standard		2.7pF	0402
C42	B3	Standard		6.2pF	0402
C43	B3	Standard		3pF	0402
C44	B3	Standard		3.6pF	0402
C45	B2	Standard		100pF	0402
C46	B2	Standard		3pF	0603
C47	B2	Standard		200pF	0603

Ref	Grid	Type	Flags	Value	Footprint
C49	A2	Standard		100nF	0603
C51	A2	Standard		1nF	0603
C52	B2	Standard		100nF	0402
C53※	B1	Debug Position		100pF	0402
C54	B1	Standard		100pF	0402
C55	B1	Standard		100pF	0603
C56	B1	Standard		100pF	0603
C57	A2	Standard		100pF	0402
C58	A2	Standard		100pF	0402
C59	A1	Standard		100pF	0402
C60	B3	Standard		22uF	0805
C61	B3	Standard		100nF	0603
C62	B3	Standard		100nF	0603
C63	B2	Standard		100nF	0603
C64	B2	Standard		100nF	0603
C65	B3	Standard		100nF	0603
C66	B3	Standard		22uF	0805
C67	B2	Standard		1uF	0603
C68	B2	Standard		18pF	0603
C69	B2	Standard		18pF	0603
C70	C2	Power Section		47nF	0603
C71	C2	Power Section		4.7uF	0603
C72	C2	Power Section		6.8uF	1206
C73	C2	Power Section		10uF	0805
C74	C2	Power Section		10uF	0805
C75	C2	Power Section		10uF	0805
C76	C3	Power Section		22uF	1206
C77	C3	Power Section		22uF	1206
C78	C3	Power Section		1uF	0603
C79	C1	Power Section		1uF	0603
C80	C2	Power Section		1uF	0603
D2	B2	Standard	|	ESD3.3V	0402
D3	C2	Standard	|	ESD3.3V	0402
F2	C3	Standard		6V/200mA	0805
F3	B2	Standard		6V/200mA	0805
F4	A2	Standard		6V/200mA	0805
F5	A1	Standard		6V/200mA	0805
FL1	B1	Standard	|	TA0970A	SMD3838–6
FL2	C1	Standard	|	TA0970A	SMD3838–6
J1	B3	Standard	|	2x20 Socket	2x20 P2.54
J2	A2	Standard	|	U.FL GNSS Ext	U.FL
J4	C2	Standard	|	USB–C	USB–C 16P
J5	C2	Standard	|	U.FL 978	U.FL

Ref	Grid	Type	Flags	Value	Footprint
J6	B1	Standard	|	U.FL 1090 Ext	U.FL
J7	B1	Standard	|	U.FL IFA	U.FL
J8	A1	Standard	|	U.FL GNSS Int	U.FL
J9※	C3	Power Section	|	MX1.25 Batt (Bottom)	MX1.25–2P
L2	A2	Standard		33nH	0402
L7	B3	Standard		6.8uH	0805
L8	C2	Standard		100nH	0402
L9	B3	Standard		7.5nH	0402
L10	B3	Standard		27nH	0402
L11	C3	Standard		6.8nH	0402
L12	C3	Standard		6.8nH	0402
L13	B3	Standard		7.5nH	0402
L14	B1	Standard		100nH	0402
L15	A1	Standard		33nH	0402
L16	C2	Power Section		1uH	SRN4018
L17	C3	Power Section		1.5uH	SRN4018
Q2	C3	Standard	|	AO3401A	SOT–23
Q3	B2	Standard	|	AO3401A	SOT–23
Q4	A2	Standard	|	AO3401A	SOT–23
Q5	A1	Standard	|	AO3401A	SOT–23
R1	A2	Standard		10k	0603
R2	A3	Standard		4.7k	0603
R3	A3	Standard		4.7k	0603
R4	A1	Standard		1k	0603
R5	A2	Standard		10k	0603
R6	B2	Standard		10k	0603
R7※	C1	No–Power Variant	✕	5.1k	0603
R8※	C1	No–Power Variant	✕	5.1k	0603
R9	B2	Standard		27R	0603
R10	B2	Standard		27R	0603
R17	C3	Standard		10k	0603
R18	B2	Standard		10k	0603
R19	B2	Standard		52.3R	0402
R21※	B2	Standard		0R	0603
R22※	B1	Standard		0R	0603
R23※	B2	Standard		0R	0603
R24※	B1	Antenna Select	✕	0R	0402
R25※	B1	Antenna Select	✕	0R	0402
R26	A2	Standard		10k	0603
R27	A2	Standard		10k	0603
R30	B2	Standard		1k	0402
R31	B2	Standard		1k	0603
R32	B2	Standard		10k	0603

Ref	Grid	Type	Flags	Value	Footprint
R33	B2	Standard		100k	0603
R34	A2	Standard		10k	0603
R35	A2	Standard		10k	0603
R36	B2	Standard		1k	0603
R37	C1	Power Section		6.8k	0603
R38	C2	Power Section		180R	0603
R39	C2	Power Section		5.62k	0603
R40	C2	Power Section		31.6k	0603
R41	C3	Power Section		470k	0603
R42	C3	Power Section		150k	0603
R43	C2	Power Section		1k	0603
R44	C1	Power Section		10k	0603
R45	C2	Power Section		10k	0603
R46	C2	Power Section		10k	0603
RT1※	C2	Power Section		10k NTC (Bottom)	0603
SW2※	A2	No Part	✕	BOOTSEL (Bottom)	Solder jmp
U1	A3	Standard	|	ME6211C33	SOT–23–5
U2	A3	Standard	|	TPS7A2033	SOT–23–5
U3	B3	Standard	|	TPS7A2033	SOT–23–5
U4	A2	Standard	|	BNO085	LGA–28
U5	A2	Standard	|	BMP388	LGA–10
U6	A3	Standard	|	QMC5883P	LGA–16
U7	A2	Standard	|	ATGM336H	LCC–18
U8	B2	Standard	|	RP2040	QFN–56
U9	B2	Standard	|	W25Q128	SOIC–8
U10	B3	Standard	|	CC1312R	QFN–48
U11	B1	Standard	|	QPL9547	DFN–8
U12	C1	Standard	|	BGA2817	SOT–363
U14	B2	Standard	|	AD8313	MSOP–8
U15	B2	Standard	|	TLV3501	SOT–23–6
U16※	B1	Antenna Select	|	XA17–G4K	SOT–363
U17※	A2	Standard	|	XA17–G4K	SOT–363
U18※	C1	Power Section	|	CH224K	ESSOP–10
U19	C2	Power Section	|	SY6970	QFN–24
U20	C3	Power Section	|	SY7069	TSOT–23–6
Y1	B1	Standard	|	12MHz	3225–4
Y2	B3	Standard	|	48MHz	3225–4
Y3	B2	Standard	|	32.768k	3215–2
ZP1※	A1	IFA Match	✕	DNP	0603
ZP2※	B1	IFA Match	✕	DNP	0603
ZS1※	A1	IFA Match		0R	0603

170 designators, sorted by reference \| Standard 131 · Power Section 28 · No–Power Variant 2 · Antenna Select 3 · IFA Match 3 · Debug Position 1 · No Part 2 \| 38 are polarised

Pilot Kit Expansion Board V4.0 Selective Placement Notes

Standard parts are fitted unconditionally and are not listed here. Every class below carries a precondition; getting it wrong breaks the function or the board — full reasoning in SELECTIVE_PLACEMENT-zh_CN.md.

No-Power Variant (2 parts, default here: DNP)

5.1k CC pull-downs, fitted on the no-power variant only. Together with U18 (CH224K) they skew the CC resistance, so PD negotiation may request 9V/12V into a board designed for 5V. Both sit inside a NO-PWR VER silkscreen box.

- R7 5.1k [DNP] Not fitted on this (powered) variant
- R8 5.1k [DNP] Not fitted on this (powered) variant

Antenna Select (3 parts, default here: See notes)

Three mutually exclusive ways to route the 1090 antenna path — fit at most one. R24+R25 together short the external antenna to the on-board IFA; U16 plus either bypass shorts a switch port.

- R24 0R [DNP] 0R bypass → hard-wires the external antenna; not fitted here
- R25 0R [DNP] 0R bypass → hard-wires the on-board IFA; not fitted here
- U16 XA17-G4K [Fit] RF switch (default option) — this is the one to fit

IFA Match (3 parts, default here: See notes)

Pi matching network for the on-board IFA. The three positions combine freely into an L or pi network and are not exclusive. The default ZS1=0R is a straight-through link, not a tuned match — final values come from a VNA sweep with the enclosure fitted.

- ZP1 DNP [DNP] Shunt, antenna side; not fitted by default
- ZP2 DNP [DNP] Shunt, radio side; not fitted by default. First sweep starts at 3.3pF
- ZS1 0R [Fit] Series position, default 0R link; start the first sweep at 3.6nH

Debug Position (1 parts, default here: Fit)

Normally fitted; removed only for one specific measurement.

- C53 100pF [Fit] Remove before sweeping the IFA through J7, otherwise the VNA sees the switch and everything behind it instead of the antenna. Refit after

Power Section (28 parts, default here: Fit)

Fitted on the powered variant only. On the no-power variant this whole group is omitted and R7/R8 are fitted instead.

- J9 MX1.25 Batt [Fit] Battery connector, on the bottom side
- RT1 10k NTC [Fit] NTC for battery temperature sensing, on the bottom side
- U18 CH224K [Fit] CH224K PD sink. Listed as discontinued on the ordering page — watch this when sourcing
- Others in this class: C70, C71, C72, C73, C74, C75, C76, C77, C78, C79, C80, L16, L17, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, U19, U20

No Part (2 parts, default here: —)

Bare copper or a solder-bridge jumper — there is no part to fit.

- ANT1 IFA Antenna [DNP] The on-board IFA — it is the PCB copper itself
- SW2 BOOTSEL [DNP] BOOTSEL solder bridge, short it with tweezers; on the bottom side

Standard (131 parts, default here: Fit)

This whole class is fitted unconditionally; only the few with an extra caveat are listed. The remaining 127 just get fitted.

- R21 0R [Fit] 0R link; open it to probe the AD8313 detector output on its own
- R22 0R [Fit] 0R link; open it to drive the 1090 switch A pin by hand
- R23 0R [Fit] 0R link; open it to drive the 1090 switch B pin by hand
- U17 XA17-G4K [Fit] GNSS-side RF switch. There is no bypass resistor on this side, so leaving it out means no signal path at all